

Bird Listening: An Application to Experience the Call and Beauty of Wild Birds

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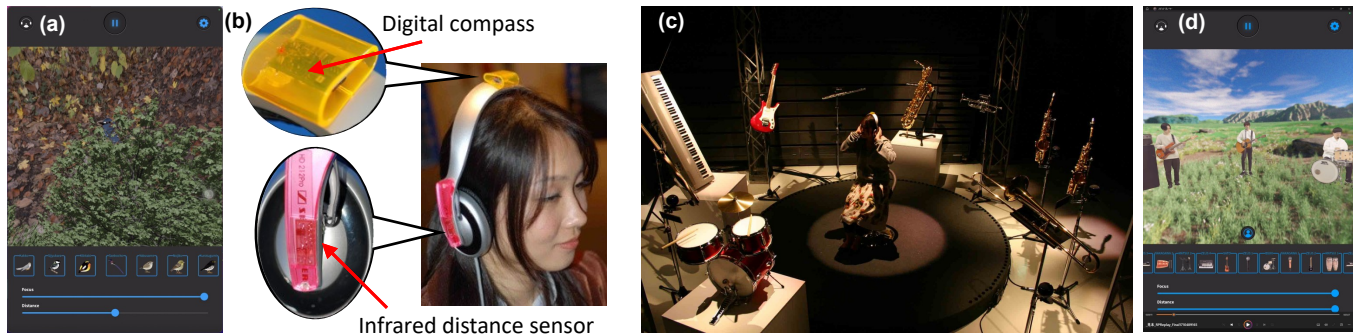


Figure 1: (a) Bird Listening, (b) Sound Scope Headphones, (c) Exhibition at SIGGRAPH2009, and (d) Sound Scope Pad.

ABSTRACT

Bird listening is an application to find and appreciate ten types of wild birds by using spatial acoustic bird sounds. When the user faces the direction of a bird's call and holds a digital device in front of him or her, the bird appears through the leaves of the tree.

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1 INTRODUCTION

Can you imagine what wild birds look like from their calls? The bird calls in this app come from the National Institute for Environmental Studies' monitoring work in the hard-to-return-to areas affected by the disaster at the Fukushima Daiichi Nuclear Power Plant. The institute set up micro-recorders in the area and recorded sounds daily for 5 minutes starting at sunrise. Bird enthusiasts can listen to the recorded data to determine the types and numbers of birds at the locations.

The app combines information from the iPad's gyro sensor and head direction detection by image processing using its front camera to achieve spatial acoustics with ordinary headphones or earphones. Previous spatial acoustic experiences require sensor-equipped headphones or a speaker array.

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Many people have trouble hearing a particular bird's call in environments where multiple birds are singing at the same time. To make listening easier, we slightly emphasize the calls that the user hears from the front. In addition, when an image from the iPad's front camera is processed to recognize the gesture of cupping a hand close to an ear to enable listening carefully, the call that the user is hearing from the front is strongly emphasized.

By holding the iPad in front of himself or herself, the user can search for wild birds by listening to their calls, and the birds appear from among the tree leaves on the screen (Fig. 1(a)). We programmatically controlled the movement of the 3D objects and enabled both sound and movement of the 3D objects by making the objects disappear in unseen areas.

2 RELATED WORK

Headphones equipped with sensors that detect the direction and position of the head are available, but because their purpose is to enhance the sense of presence by fixing the virtual sound source position [Goudeseune and Kaczmarek 2001; Oliver and Gerhard 2004; Wu et al. 1997], they cannot be used to emphasize a specific part. Thus, these headphones make it difficult to selectively listen to a certain instrument located near other instruments.

A spatial audio system makes it possible to change the volume of an instrument by moving the position of the listener's avatar and each instrumental part [Pachet and Delerue 1998, 2000]. However, beginners can have difficulty properly adjusting the mixing of each part because the process is complicated; when a solo begins, the part must be placed in the center localization, and when the piece returns to the accompaniment, the part must be placed far away.

We previously proposed Sound Scope Headphones that enable users listening to a song to focus on the sound of a specific part (Fig. 1(b)) [Hamanaka 2006; Hamanaka and Lee 2006, 2013]. By turning their head to the left or right, users can follow the sound

of specific instruments on the respective sides. Moving their palms close to their ears in a “listening” pose enables them to focus on specific parts. The Sound Scope Headphones use a digital compass that detects the orientation of the face and distance sensors that measure the distance between the hand and ear.

Figure 1(c) shows our exhibition at ACM SIGGRAPH2009 Emerging Technologies [Hamanaka and Lee 2009]. When face-to-face exhibitions were difficult to hold due to social distancing restrictions during the COVID-19 Pandemic, we investigated implementing similar functions in an app called Sound Scope Phone [Hamanaka 2022b] and Sound Scope Pad [Hamanaka 2022a] so that a wider audience could experience our demonstration by using a tablet.

3 SOUND SCOPE PAD

By using the Sound Scope Pad application installed on an iPad, users can experience the following.

3.1 VR Concert Experience

Performers play while standing around the user. The user can see three performers at a time in front of them on the iPad screen. If the user points the iPad to the left, the performer on the left will appear, and if the user points it to the right, the performer on the right will appear (Fig. 1(d)).

3.2 Spatial Acoustic Experience

If the user holds the iPad in front, they will see the application tracking their face on the display. The user can turn their head left or right to follow the sound of the performers on their respective sides. Likewise, if the user turns around while holding the iPad, they will be able to face the performers behind them.

3.3 Active Music Listening Experience

The app provides an active music listening experience by enabling the user to focus on certain performers that they want to hear. If the user turns to face the performer of their choice, it will sound as though the performer is directly in front of the user. The user can move their left or right palm closer or farther away from their ear to adjust the range of the performance that they want to focus on. The user can highlight one or multiple parts of the performance, such as the horn or rhythm section. The user can check the direction of the performer and the direction they are facing on the position display.

4 IMPLEMENTATION OF BIRD LISTENING

Bird Listening has implemented the following two new functionalities in addition to those of the Sound Scope Pad.

4.1 Combination of 3D Objects and Videos

The Sound Scope Pad was implemented in Swift [Apple 2024], Apple’s programming language, and was able to display 10 players by playing 10 videos simultaneously in a 360-degree space. The wild birds were created in 3D computer graphics (3DCG) because there was no video, although calling sound data recorded by micro-recorders was available. We tried to run the bird animation we created on the Blender [Foundation 2024] CG software in Swift, but the iPad became overloaded and slow, and the 3D objects were not

displayed. Therefore, we decided to play the animation as a video. On the other hand, the 3D object was imported into Swift to make the tree where the bird stops appear three-dimensional.

4.2 Reuse of Object Names

In the development environment on the iPad operating system (OS), the OS is managed by memory, which is usually automatically released when objects are erased. In other words, there is no way to release memory directly from the program side. However, when a COLLADA format file [Group 2024] created by Blender or other software is loaded, memory is not released even if the object is erased. Multiple contents included in Sound Scope Pad, such as Bird Listening, led to insufficient memory when switching to another content, as the memory of the previous content was not released. To solve this problem, we implemented a common name for the 3D scenes and video objects used in each content and reused them, thereby addressing the memory shortage problem.

5 CONCLUSION

Bird Listening will be available for download in the next update of Sound Scope Pad. The application can be downloaded and introductory video can be viewed at the following URL:

<https://gttm.jp/hamanaka/en/soundscopepad/>.

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